

Incompatible counting and growth assertions in Conjecture

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The conjecture asserts both $a_{r,s} = D(r,s)$, the Delannoy numbers, and $a_{m,m} \sim c3^m m^{-1/2}$. These assertions are incompatible. The contradiction does not depend on any interpretation of the Bailey sliding construction: no array can satisfy the two asserted properties together.

A lower bound for all diagonal entries. The Delannoy numbers are characterized by

$$D(0,s) = D(r,0) = 1, \quad D(r+1,s+1) = D(r,s+1) + D(r+1,s) + D(r,s).$$

They count paths using steps $(1,0), (0,1), (1,1)$, by conditioning on the last step. They are nondecreasing in each coordinate. Set $x = D(r,s)$. The recurrence and this monotonicity give

$$D(r+1,s+1) \geq 3x, \quad D(r+1,s+2) \geq 5x, \quad D(r+2,s+1) \geq 5x.$$

Thus $D(r+2,s+2) \geq 13D(r,s)$, and induction gives

$$D(2n,2n) \geq 13^n \quad (n \geq 0).$$

In fact, the diagonal is not even $O(3^m)$. For a completely elementary estimate, induction on n proves

$$(4n+9)9^n \leq 9 \cdot 13^n.$$

The induction step follows from $9(4n+13) \leq 13(4n+9)$. If some integers $C, N \geq 0$ satisfied $D(m,m) \leq C3^m$ for all $m \geq N$, take $n = 9C + N + 1$ and $m = 2n$. Then

$$(4n+9)9^n \leq 9 \cdot 13^n \leq 9C9^n,$$

so cancellation of the positive number 9^n gives $4n+9 \leq 9C$, contrary to the chosen n .

Why the stated asymptotic is excluded. The usual meaning of $a_{m,m} \sim c3^m/\sqrt{m}$ has a finite positive constant c . For all sufficiently large m , it entails $a_{m,m} \leq 2c3^m/\sqrt{m} \leq 2c3^m$. Choose an integer $C \geq 2c$ and enlarge the integer threshold N to at least one. This gives precisely the eventual bound just disproved. A nonpositive c cannot supply an asymptotic equivalent of the positive Delannoy diagonal either.

Formal verification and scope. Lean defines the full two-variable sequence by the displayed terminating recurrence, and proves uniqueness among every array satisfying the two boundary conditions and the recurrence. The monotonicity, block-growth estimate, and diagonal lower bound are theorems for arbitrary natural indices. There is no extrapolation from a finite table.

The theorem `no_eventual_three_power_bound` negates the existence of every integer bound C and every threshold N . The theorem `conjecture7689_counterexample` then negates, for any array a , the conjunction of equality to Delannoy numbers and this necessary eventual bound. The preceding paragraph supplies explicitly the elementary real-asymptotic implication to that integer statement; the Lean file formalizes its integer consequence rather than introducing a new or nonstandard definition of asymptotic equivalence. The other assertion about product-side factors is not needed to refute the conjunction.

The program uses Lean 4.19.0 and `Std` only, with no admitted lemmas, custom axioms, or native evaluation proofs.